

Alexis DHERMY

Applicant for M.S. Program in Artificial Intelligence – South Korea (Spring 2027)

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EDUCATION

EPF Engineering School — Paris-Cachan, France

Sep 2021 – Aug 2026

Master of Science in Engineering (French Diplôme d'Ingénieur)

Major: Computer Science | Track: Artificial Intelligence

- GPA: **15.87/20** | Rank: **Top 3% of cohort** (Y4: 16/276, Y3: 6/206, Y2: 4/178, Y1: 4/217)
- Thesis: LLM-Based RAG Architecture for Enterprise Systems (*Viridien*) | Advisor: Prof. Zehira Haddad
- 2-year intensive preparatory cycle (Mathematics & Physics), 1-year general engineering, 2-year specialization (CS & AI)
- Relevant Coursework: Generative AI, Deep Learning, Machine Learning, Probability & Statistics, Linear Algebra

Sungkyunkwan University — Seoul, South Korea

Sep 2024 – Dec 2024

Exchange Semester

Department: Applied Artificial Intelligence

- GPA: **4.10/4.5**
- Relevant Coursework: Reinforcement Learning, Natural Language Processing, Large Language Models, Machine Learning, AI Search Algorithms, Data Science

HONORS & AWARDS

National Engineering Entrance Merit Distinction — Concours Avenir (*Grand Classé*)

May 2021

- Selected in the top tier of 10,000+ candidates for academic excellence, in one of France's most competitive post-high school engineering entrance exams, earning full written test exemption.

RESEARCH INTERESTS

- LLMs, Reasoning Models, AI Agents, Efficient Inference
- Reinforcement Learning, Embodied AI, Robotics, World Models

ACADEMIC PROJECTS

Music Recommendation Using NLP [[Code & Report](#)] — Sungkyunkwan University

Nov 2024 – Dec 2024

- Developed an end-to-end NLP pipeline (NLTK, SpaCy, BERT) to perform tokenization, POS tagging, lemmatization, and sentiment analysis on top Spotify tracks across France, Spain, and Korea.
- Built a cross-cultural recommendation engine that maps semantic and emotional features to recommend French/Spanish tracks based on a target Korean song.

Teaching a Robot to Walk with RL [[Presentation](#)] — Sungkyunkwan University

Nov 2024 – Dec 2024

- Trained a 2D bipedal robot in OpenAI Gym (BipedalWalker-v3) to navigate complex terrain through deep reinforcement learning.
- Designed custom reward shaping combining speed, distance, and joint stability with a step-based delay threshold, driving a 3x increase in distance and velocity over baseline Advantage Actor-Critic (A2C).
- Implemented Proximal Policy Optimization (PPO) to mitigate policy collapse, boosting average episode moving distance by 132%.

PROFESSIONAL EXPERIENCE

AI Software Engineer — Viridien (R&D Department – Seismic Imaging), Massy, France

Feb 2026 – Jul 2026

Master's Thesis Internship

- Conducted applied AI research to architect an enterprise advanced RAG system (LlamaIndex, Python), parsing 80k+ geo-physical technical documents and DSL scripts to boost response accuracy from 54% to 90%.
- Built a multi-stage ingestion pipeline featuring Docling parsing, automated cleaning, hierarchical chunking, Anthropic's Contextual Retrieval, and LLM-generated DSL summaries.
- Optimized the retrieval engine using metadata filtering, hybrid search (dense + BM25), parent-child linking, and reranking.
- Benchmarked configurations (chunk sizes, embedding models, fusion strategies, LLMs) to double retrieval efficiency.
- Engineered a Model Context Protocol (FastMCP) server exposing 2 retrieval tools and 3 instructional prompts for LLM agents.
- Deployed into the official enterprise AI platform using Docker, Qdrant, MongoDB, and Ollama.

Corporate Academic Project

- Deployed autonomous visitor onboarding and multimodal interaction flows on the Pepper humanoid robot by engineering native Android applications (Kotlin, SoftBank QiSDK).
- Reverse-engineered legacy robot architecture and internal systems to build a real-time engagement pipeline combining sensor-based presence detection, voice commands, and tablet UI.
- Modernized the robot's dialogue pipeline by integrating Mistral LLM APIs, upgrading basic command handling into dynamic, context-aware conversational flows.

Backend & Cloud Developer — EPF Projets, Cachan, France

Jul 2025 – Oct 2025

Freelance

- Engineered the backend for "Carbon Track", a carbon-footprint mobile application, building REST APIs (Node.js, Express) connected to ADEME and Open Food Facts environmental databases.
- Designed a hybrid database & serverless backend combining Firebase (Cloud Functions, Firestore, Auth) with local SQLite for offline data persistence.
- Automated mobile build and release pipelines using the Expo/EAS CI/CD stack to test, package, and deploy the native application onto the Google Play Store.

TECHNICAL SKILLS

- **Programming:** Python, Java, SQL, Bash
- **AI Frameworks:** PyTorch, Hugging Face, Gymnasium (OpenAI Gym), LangChain, LlamaIndex
- **Transformer & LLM Architectures:** Attention Mechanisms, KV-Caching, Fine-Tuning (SFT, LoRA)
- **Reinforcement Learning:** Value-Based (Q-Learning, DQN), Policy Gradients & Actor-Critic (A2C, PPO), Reward Shaping
- **Generative AI:** AI Agents, MCP, Advanced RAG (Dense, BM25, SPLADE, ColBERT, Reranking), Vector DBs (Qdrant), HNSW
- **Infrastructure & Tools:** Linux, Docker, Git, CI/CD, Cloud (AWS, GCP, Azure)
- **Languages:** English (Professional, TOEIC 895), French (Native), Korean (Basic, SKKU Coursework)